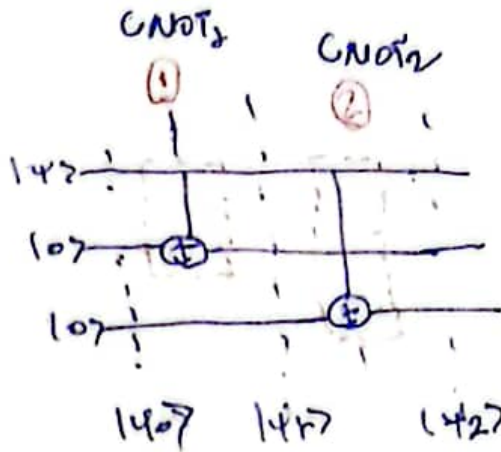


Problem-2

n) Given

Circuit



$$|\psi\rangle = a|0\rangle + b|1\rangle$$

$$CNOT_1 = P_0 \otimes I \otimes I + P_1 \otimes X \otimes I$$

$$CNOT_2 = P_0 \otimes I \otimes I + P_1 \otimes I \otimes X$$

Solution

Verify this circuit indeed implements such an encoding.

where

$$P_0 = |0\rangle\langle 0|$$

$$P_1 = |1\rangle\langle 1|$$

Therefore the state must be Krone-product

$$\begin{aligned} |\psi_0\rangle &= |\psi\rangle \otimes |0\rangle \otimes |0\rangle \\ &= (a|0\rangle + b|1\rangle) |00\rangle \\ &= \underline{a|000\rangle + b|100\rangle} \end{aligned}$$

1st - Apply the control between the first qubit and second qubit

$$\begin{aligned} |\psi_1\rangle &= CNOT_1(|\psi_0\rangle) \\ |\psi_2\rangle &= CNOT_2(a|000\rangle + b|100\rangle) \end{aligned}$$

Control = qubit-1
Target = qubit-2

$$= \underline{a|000\rangle + b|110\rangle}$$

2nd Apply the control gate between the second qubit and third qubit

Control = qubit-2
Target = qubit-3

↓

$$\begin{aligned}
 | \psi \rangle &= CNOT_2(| \psi \rangle) \\
 &= CNOT_2(| 0000 \rangle + | 1111 \rangle) \\
 &= | 0000 \rangle + | 1111 \rangle
 \end{aligned}$$

2) Solution

The encoder

$$| \psi \rangle = | 0 \rangle + | 1 \rangle \rightarrow$$

$$| \psi \rangle_L = | 0000 \rangle + | 1111 \rangle$$

It protects against bit flip error
 X_1, X_2, X_3
 because a single "X" changes one
 bit and majority voting can
 detect and correct it.

What a phase flip error

does

$$Z|0\rangle = |0\rangle$$

$$Z|1\rangle = -|1\rangle$$

unlike example

$$Z(| 0000 \rangle + | 1111 \rangle) = | 0000 \rangle - | 1111 \rangle$$

Note: not an orthogonal error state - F+FI
 just the original logical state with
 relative phase change.
Required

3) Given

$$\Sigma_L(| \psi \rangle) = (1-p)| \psi \rangle_L + pX| \psi \rangle_L \quad P_{\text{err}}(| \psi \rangle, \Sigma_L(| \psi \rangle)) = ?$$

Solution

The fidelity between a pure state $| \psi \rangle$ and arbitrary map
 quantum state $\Sigma_L(| \psi \rangle)$

$$F(| \psi \rangle, \Sigma_L(| \psi \rangle)) = \langle \psi | \Sigma_L(| \psi \rangle) | \psi \rangle$$

$$F(| \psi \rangle, \Sigma_L(| \psi \rangle)) = \langle \psi | [(1-p)| \psi \rangle_L + pX| \psi \rangle_L] | \psi \rangle$$

$$= (1-p) \underbrace{\langle \psi | \psi \rangle}_{=1} \underbrace{\langle \psi | \psi \rangle}_{=1} + p \underbrace{\langle \psi | X | \psi \rangle}_{=0} \langle \psi | \psi \rangle$$

=

$$= \langle \psi | X | \psi \rangle^2$$

$$= (\lambda - 1) \cdot \lambda - \lambda + 1 \cdot \langle \psi | \psi \rangle^2$$

For the minimum fidelity occurs at $\langle \psi | \psi \rangle = 0$

$$F_{\min}(\langle \psi |, \hat{Z}_2(\langle \psi |)) = (\lambda - 1) \cdot \lambda - \lambda + 1 \cdot 0 \\ = \underline{\underline{\lambda - 1}}$$

b) Solution

$$\hat{Z}_3(\langle \psi |) = [(\lambda - 1)^3 + 3\lambda(\lambda - 1)^2] \langle \psi | \psi \rangle$$

$$F(\langle \psi |, \hat{Z}_3(\langle \psi |)) = \langle \psi | \hat{Z}_3(\langle \psi |) \langle \psi |$$

$$= \langle \psi | [(\lambda - 1)^3 + 3\lambda(\lambda - 1)^2] \langle \psi | \psi \rangle \langle \psi |$$

$$= (\lambda - 1)^3 + 3\lambda(\lambda - 1)^2 \underbrace{\langle \psi | \psi \rangle}_{=1} \underbrace{\langle \psi | \psi \rangle}_{=1}$$

$$= \underline{\underline{(\lambda - 1)^3 + 3\lambda(\lambda - 1)^2}}$$

c) Solution

from (a) minimum fidelity without error correction

$$F_{\text{uncorrected}} = \lambda - 1$$

from (b) the lower bound on the fidelity after error correction is

$$F_{\text{corrected}} \geq (\lambda - 1)^3 + 3\lambda(\lambda - 1)^2$$

Error correction improves the fidelity as occurs

$$(\lambda - 1)^3$$

$$F_{\text{corrected}} > F_{\text{uncorrected}}$$

3)

$$(2-p)^3 + 3p(2-p)^2 > 2-p$$

$$= (2-p)^2 (2+2p)$$

$$(2-p)^2 (2+2p) > 2-p, \quad \text{where} \quad (2-p)^2 (2+2p) = 2 - 3p^2 + 2p^3$$

$$2 - 3p^2 + 2p^3 > 2-p$$

$$-3p^2 + 2p^3 + p > 0,$$

$$p(2p^2 - 3p + 1) > 0$$

$$(2p^2 - 3p + 1) > 0, \quad \text{where} \quad 2p^2 - 3p + 1 = (2p-1)(p-1)$$

$$(2p-1)(p-1) > 0$$

Determine the valid range of probability

physical probabilities $0 \leq p \leq 1$.

$(2p-1)(p-1) > 0$, the equation valid when

$$0 < p < \frac{1}{2}$$

$$\underline{\underline{II}}$$

4) a) solution parameters of the two collections

① The two two collections encode 2 logical qubits

② It uses 9 physical qubits

③ It can correct any single-qubit error so for distance $d=3$

$$[[n, k, d]] = [[9, 2, 3]]$$

4)

b) Show that

$$\left. \begin{array}{l} X_1 X_2 X_3 X_4 X_5 X_6 \\ X_4 X_5 X_6 X_7 X_8 X_9 \end{array} \right\} \text{phase-flip errors}$$

phase-flip protection

$$|0\rangle_L \equiv \frac{|000\rangle_L + |111\rangle_L}{\sqrt{2}}, \quad |1\rangle_L \equiv \frac{|000\rangle_L - |111\rangle_L}{\sqrt{2}}$$

$$\textcircled{*} |000\rangle_L = |0\rangle_L^{(1)} |0\rangle_L^{(2)} |0\rangle_L^{(3)}, \quad |111\rangle_L = |0\rangle_L^{(1)} |1\rangle_L^{(2)} |1\rangle_L^{(3)}$$

The three-block phase-flip code uses the standard three-qubit phase-flip stabilizers

$$X_1^L X_2^L \text{ and } X_2^L X_3^L$$

$$X_1^L = X_1 X_2 X_3$$

$$X_2^L = X_4 X_5 X_6$$

$$X_3^L = X_7 X_8 X_9$$

then the stabilizers become

$$X_1^L X_2^L = X_1 X_2 X_3 X_4 X_5 X_6$$

$$X_2^L X_3^L = X_4 X_5 X_6 X_7 X_8 X_9$$

phase-flip syndrome measures

$$X_1 X_2 X_3 X_4 X_5 X_6$$

$$X_4 X_5 X_6 X_7 X_8 X_9$$

$$\underline{\underline{\quad \quad \quad / \quad / \quad / \quad \quad \quad}} \quad \underline{\underline{\quad \quad \quad / \quad / \quad / \quad \quad \quad}}$$

c) Solution

decompose Y_i

$$Y_i = X_i Z_i$$

⊙ A Y_i error is detected as simultaneous bit and phase-flip syndromes and corrected by applying X_i and Z_i or equivalently Y_i to remove the qubit.

4) Solution

The depolarizing channel acts on a single qubit as

$$Z(\rho) = (1-p)\rho + p\frac{I}{2}, \quad \frac{I}{2} = \frac{1}{4}(\rho + X\rho X + Y\rho Y + Z\rho Z)$$

$$Z(\rho) = (1-p)\rho + \frac{p}{4}(\rho + X\rho X + Y\rho Y + Z\rho Z)$$

$$= (1-p)\rho + \frac{p}{4}\rho + \frac{p}{4}(X\rho X + Y\rho Y + Z\rho Z)$$

$$= \left(1-p + \frac{p}{4}\right)\rho + \frac{p}{4}(X\rho X + Y\rho Y + Z\rho Z)$$

$$= \left(1 - \frac{3p}{4}\right)\rho + \frac{p}{4}(X\rho X + Y\rho Y + Z\rho Z)$$

⊙ The depolarizing channel written as a probabilistic mixture of Pauli errors X, Y, Z

⊙ It is equivalent to applying one of the discrete Pauli errors $\{I, X, Y, Z\}$ with certain probabilities

⊙ The Shor code can correct any single-qubit Pauli error, including X, Y, Z

⊙ The Shor code can protect against depolarizing noise because the depolarizing channel can be discretized into correctable single-qubit Pauli errors.